High Resolution Wake Capturing Methodology for Hovering Rotors





Karthikeyan Duraisamy*
Research Scientist

James D. Baeder
Associate Professor

Alfred Gessow Rotorcraft Center
University of Maryland at College Park
College Park, MD

A high-resolution Reynolds-averaged Navier–Stokes (RANS) solver is applied to study the evolution of tip vortices from rotary blades. The numerical error is reduced by using high-order accurate schemes on appropriately refined meshes. To better resolve the vortex evolution, the equations were solved on multiple overset grids that ensured adequate resolution in an efficient manner. For the RANS closure, a one equation wall-based turbulence model was used with a correction to the production term to account for the stabilizing effects of rotation in the core of the tip vortex. While experimental comparison of the computed vortex structure beyond a few chord lengths downstream of the trailing edge is lacking in the literature, reasonable validation of the vortex velocity profiles is demonstrated up to a distance of 50 chord lengths of evolution for a single-bladed rotor. For the two-bladed rotor case, the tip vortex could be tracked up to two rotor revolutions with minimal diffusion. The accuracy of the computed blade pressures and vortex trajectories confirm that the inflow distribution and blade-vortex interaction are represented correctly. The accuracy achieved in the validation studies establishes the viability of the methodology as a reliable tool that can be used to predict vortex evolution and the aerodynamic performance of hovering rotors.

Nomenclature

C_p	pressure coefficient $\frac{p-p_\infty}{0.5\rho_\infty(U_{\text{tip}}y/R)^2}$
C_T	rotor thrust coefficient
c, R	blade chord and radius (m)
q	non-dimensionalized invariant of the velocity gradient tensor $\frac{\partial u_i}{\partial x_j}\frac{\partial u_j}{\partial x_i}\frac{c^2}{U_{\text{tip}}^2}$
Re	Reynolds number based on chord and tip speed
U_{tip}, M_{tip}	tip speed (m/s) and Mach number
y/R	non-dimensional spanwise coordinate
θ_0	collective pitch (degrees)
ν	kinematic viscosity (m^2/s)
ρ_∞, p_∞	freestream density (kg/m^3) and pressure (N/m^2)
τ_w	wall shear stress (N/m^2)
ψ	wake age (degrees)

Introduction

The wake generated by a helicopter blade is characterized by concentrated tip vortices. These tip vortices are typically strong enough to persist for many rotor revolutions, and the resulting flow field is highly unsteady. The formation and evolution of the tip vortices is highly dependent on the blade loading. In turn, the velocities induced by the wake significantly affect the blade loading and the flapping motion of the blades. Therefore, accurate modeling of the wake and its interaction with the rotor system is critical to the successful prediction of overall rotor aerodynamics and rotor performance.

*Corresponding author; email: dkarthik@glue.umd.edu.
Manuscript received September 2005; accepted June 2006.

Traditional rotor wake modeling efforts have primarily involved the use of free-wake methods (Refs. 1–6). These are potential flow based methods, which are relatively inexpensive and can often provide useful predictions and insight, but rely on empirical inputs. For instance, the free wake model of Bhagwat and Leishman (Ref. 4) assumes a vortex filament representation of the wake, and hence, relies on empiricism for the vortex release locations, initial vortex strengths, core structure, and viscous and turbulent diffusion.

The Reynolds-averaged Navier–Stokes (RANS) equations currently represent the highest level of fluid dynamic approximation that can be utilized in the complete numerical simulation of flow at Reynolds numbers relevant to fixed- and rotary-wing applications. With the development of computational power over the past two decades, these simulations have been applied in the solution of practical rotor problems. Most of this research has, however, concentrated on on-surface load distributions. For rotary-wings, most comprehensive analyses use a prescribed/Lagrangian wake model to simulate the tip vortex induced inflow (Ref. 7). More recently, overset meshes (Refs. 8–10) have been used to "capture" the wake instead of modeling the same. Even though these calculations showed reasonable comparison of integrated coefficients (and sometimes surface pressures) with experimental/flight test data, only the "overall" effects of the wake were captured because of excessive numerical dissipation. Thus, the internal structure of the tip vortex is not addressed. The authors are not aware of any publication in which a detailed comparison of the vortex structure with experimental data has been attempted beyond a downstream distance of three chords from the trailing edge.

The primary objective of this work is to use improved algorithms to reduce numerical and turbulence modeling errors, and to evaluate the methodology for accurately calculating the vortex wake structure. An additional objective is to keep the computational cost to a minimum. The methodology will be validated with the following experimental data sets:

1) The single-bladed hovering rotor of Martin et al. (Ref. 11): Detailed measurements of the vortex structure are available up to a wake age of one rotor revolution.

2) The two-bladed Caradonna–Tung rotor system (Ref. 12): Comprehensive measurements of blade surface pressure and vortex trajectory are available for a range of thrust conditions.

Background

A successful methodology for the detailed prediction of vortex structures has proved to be elusive because of a variety of issues. These include numerical errors, grid requirements, and RANS closure.

Numerical errors

Even if one assumes that the governing equations exactly model the physical phenomena, the numerical solution is bound to generate errors. In the case of time dependent partial differential equations like the Euler and Navier–Stokes equations, discretization errors can be viewed (in a linear sense) as arising from numerical dispersion and numerical dissipation (Ref. 13). Numerical dispersion causes waves of different wave lengths to propagate at different speeds, and can result in spurious oscillations in high gradient regions and shocks. Numerical dissipation acts as an artifical diffusion mechanism and is the primary factor that causes inaccuracies in most tip vortex calculations. Unlike non-linear features like shocks, tip vortices have no re-steepening mechanisms, and hence artificial decay of the vortex will result in an irrecoverable loss of information. Typically, the use of high-order accurate schemes is essential to reduce numerical errors in tip vortex simulations (Ref. 14).

Grid requirements

The aforementioned numerical errors can be reduced by using high-order accurate numerical schemes in conjunction with adequate mesh resolution. In RANS-based tip vortex simulations, two different types of grid spacings are required:

1) In the near-wall region, the normal distance between a point on the surface and the first off-surface point should be roughly one wall-unit ($\nu/\sqrt{\tau_w/\rho}$), and the stretching in the normal direction should be small. This is required to accurately represent the wall shear stress in the laminar sub-layer and the buffer layers (Ref. 15). For a blade at a chord based $Re \approx 1 \times 10^6$, a wall unit would correspond to a physical spacing of $\approx 5 \times 10^{-6} c$. In addition, the spacing in the non-normal direction can be orders of magnitude larger.

2) In the wake region, a fine mesh spacing is required to resolve the tip vortex. Based on inviscid computations on an isolated tip vortex, it is estimated that (Ref. 14) at least 15 points are required across the vortical core. In addition, studies (Refs. 14, 16) have also hinted at the importance of maintaining reasonable cell aspect ratios (to account for features that are non-orthogonal to grid planes), thus imposing stringent resolution requirements in all three directions. In practical computations, a compromise has to be achieved between maintaining the required resolution, while keeping the overall computational cost manageable.

RANS closure

The averaging of the Navier–Stokes equations introduces the need for turbulence models. The most popularly used closure models are the linear eddy viscosity models like $k - \varepsilon$ (Ref. 17), Spalart–Allmaras (Ref. 18), etc. In this approach, the Reynolds stress is related to the mean strain rates assuming isotropy. Though simple, this has turned out to be the only feasible solution to turbulence modeling for complex flows and configurations. One of the deficiencies of linear eddy viscosity closures is that these models cannot implicitly account for effects like streamline curvature and solid body rotation, which are known to be significant in tip vortex flows. On the other hand, more complex RANS closures like the so-called Second Moment Closure (SMC) (Ref. 19) models can naturally account for these effects because Reynolds stresses are inherently related to flow gradients. Even though the SMC models appear to be physically appealing and have been shown to successfully model representative problems (Ref. 20), consistent superiority of these models has not been established for complex problems (Ref. 21). In addition, practical implementation of these models is known to be difficult because of their inherent complexity and lack of numerical robustness (Ref. 22).

It has to be mentioned that in addition to the aforementioned issues, development and validation of computational methodologies suffer from a dearth of reliable experimental reference data on tip vortex evolution. While the far-field vortex measurements are affected by vortex instabilities and aperiodicity (Refs. 11, 23, 24), near-field studies are complicated by the fact that the flow gradients and turbulence intensities are large.

In the present work, the use of proper grid resolution (by means of grid clustering and overset meshes) and high-order accurate numerical schemes are used to minimize numerical errors. A simple correction to the production term of the turbulence model to account for rotational effects is seen to provide reliable predictions.

Computational Methodology

Computations were performed using a modified version of the compressible RANS code TURNS (Ref. 25). The inviscid terms were computed using the third/fifth order WENO schemes (Ref. 26) with Roe's flux difference splitting. The viscous terms were computed using the second/fourth order central differencing. The Spalart–Allmaras (Ref. 18) turbulence model was used to close the RANS equations. The equations were solved in the rotating reference frame, and hence a steady-state solution was sought. The convergence of the flow solution was assessed by at least 3.5 orders of magnitude of reduction in flow solver and turbulence residuals, after which the run was stopped when the total thrust started to fluctuate (at a small amplitude) around a mean state.

Many experimental (Refs. 23, 27) and analytical (Ref. 28) studies on tip vortices have reported largely reduced turbulence levels in the vortical core. This has been attributed to the near solid body rotation that prevails in the core. It is known (Ref. 20) that linear eddy viscosity closures cannot model this behavior implicitly. To accurately represent the physics, one has to ensure that the turbulence model does indeed replicate this phenomena. The Spalart–Allmaras model bases the production term on the magnitude of the vorticity vector. This is known to perform well for wall-bounded flows. However, the core region of the tip vortex is highly vortical and axisymmetrically strained, but as mentioned earlier suppresses turbulence. The S–A model uses a production term, which in the wake region reduces to

$$P = c_{b1} S \nu_t, \tag{1}$$

where $c_{b1} = 0.1355$ and ν_t is the eddy-viscosity function. Computations were performed with $S = |\omega|$ and $S = |\omega| + 2\min(0, |D| - |\omega|)$, where $|\omega|$ and $|D|$ are the magnitudes of the vorticity and strain tensors, respectively. The former was suggested in the original paper (Ref. 18); the latter correction was first used in Ref. 29 and attempts to suppress the turbulent production in regions where the vorticity is much larger than the strain rate (like tip vortices). The modification will be passive in thin shear layers where the magnitudes of the strain rate and vorticity are roughly equivalent.

Table 1. Experimental test conditions for single-bladed rotor (Ref. 11)

Aspect ratio	9.12
Tip Mach number	0.26
Chord Reynolds number	272,000
Collective pitch	4.5°
Root cut-out	20%

Single-Bladed Hovering Rotor

Experimental configuration

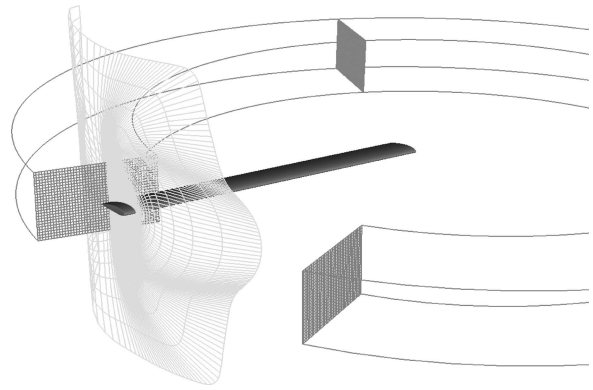
The test case chosen is the experiment of Martin et al. (Ref. 11). A single untwisted blade with rectangular planform (with a square tip shape), and NACA 2415 cross-section was used. The relevant test case details are given in Table 1. As mentioned earlier, accurate measurement of the vortex velocity profiles especially in the far-field is difficult because of the uncertainties associated with the aperiodicity of the vortex (Ref. 30). Hence, the measured profiles have to be corrected. The present test case was corrected by Martin et al. (Ref. 11) using the method described in Ref. 31.

Grid system

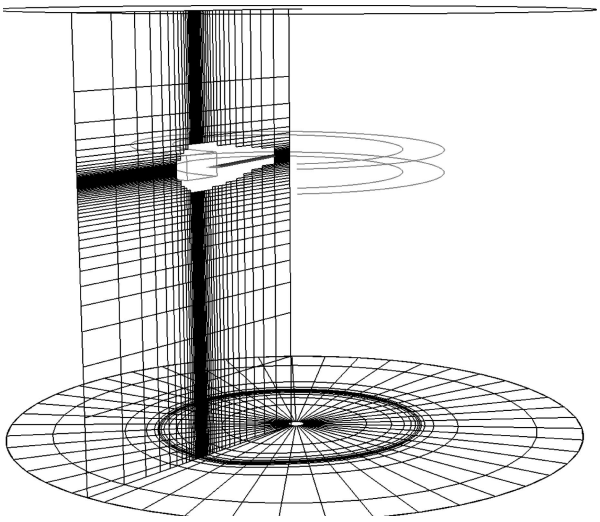
For the computations, a three-grid system was used. For the blade grid, a body-conforming C–O mesh of dimension $161 \times 55 \times 48$ ($\approx$0.43 million points) in the streamwise, spanwise and normal directions, respectively, was used. Since the flow field of a hovering rotor is cylindrical, the spanwise sections of this mesh are roughly aligned with the local flow direction in the wake region. Compared to a non-rotated mesh, the computed vortex profiles were found to be much less dissipative as a result of rough alignment with the vortex axis. The grid extended to a normal distance of slightly more than 2 chord lengths, and the normal spacing at the wall is $1 \times 10^{-5}c$ ($<$1 wall unit).

A refined overset grid of dimension $135 \times 176 \times 101$ ($\approx$2.4 million points) was used to resolve the tip vortex (Fig. 1). This grid starts at the trailing edge of the rotor and extends to a wake age of around 330°. Each streamwise section (176×101) of this grid is equi-spaced, consisting of square cells of side $0.01c$. In the axial direction, hyperbolic tangent stretching was used. In the high gradient regions, ($0° < \psi \leq 45°$) the axial spacing ranged from $0.5° \leq \Delta\psi \leq 2°$. Beyond this region, the spacing ranged from $2° \leq \Delta\psi \leq 6°$. The placement and alignment of this grid was based on the predicted wake geometry given by Landgrebe's model (Ref. 32).

To account for the far-field boundaries, a cylindrical background mesh of size $181 \times 50 \times 60$ ($\approx$0.54 million points) in the streamwise, spanwise and normal directions, respectively, was utilized as shown in Fig. 1(b). The radius of the outer boundary of this grid is twice the rotor radius, and the top and bottom boundaries are one rotor radius and two rotor radii away from the blade surface, respectively. Figure 1(b) also shows that the part of the background grid that overlaps with the blade grid is hole-cut. At the outer boundaries of the background grid, the point-sink/momentum theory characteristic boundary condition (Ref. 25) was used. The outer boundaries in the current calculations are relatively close to the blade surface as compared to that normally used in the literature. This could potentially affect the rotor loading. For instance, Ahmad and Strawn (Ref. 9) show that for a 4-bladed rotor case, the solution was insensitive to change in boundary location for boundary distances beyond three radii from the blade surface. Shorter distances changed the computed thrust levels by a few percent. It has to be mentioned however, that the thrust levels in the present calculations are very low compared to the one found in the literature, and hence, the velocities induced at the boundaries would be much smaller.



(a) Spanwise section of blade grid and streamwise sections of vortex grid



(b) Background and vortex grids

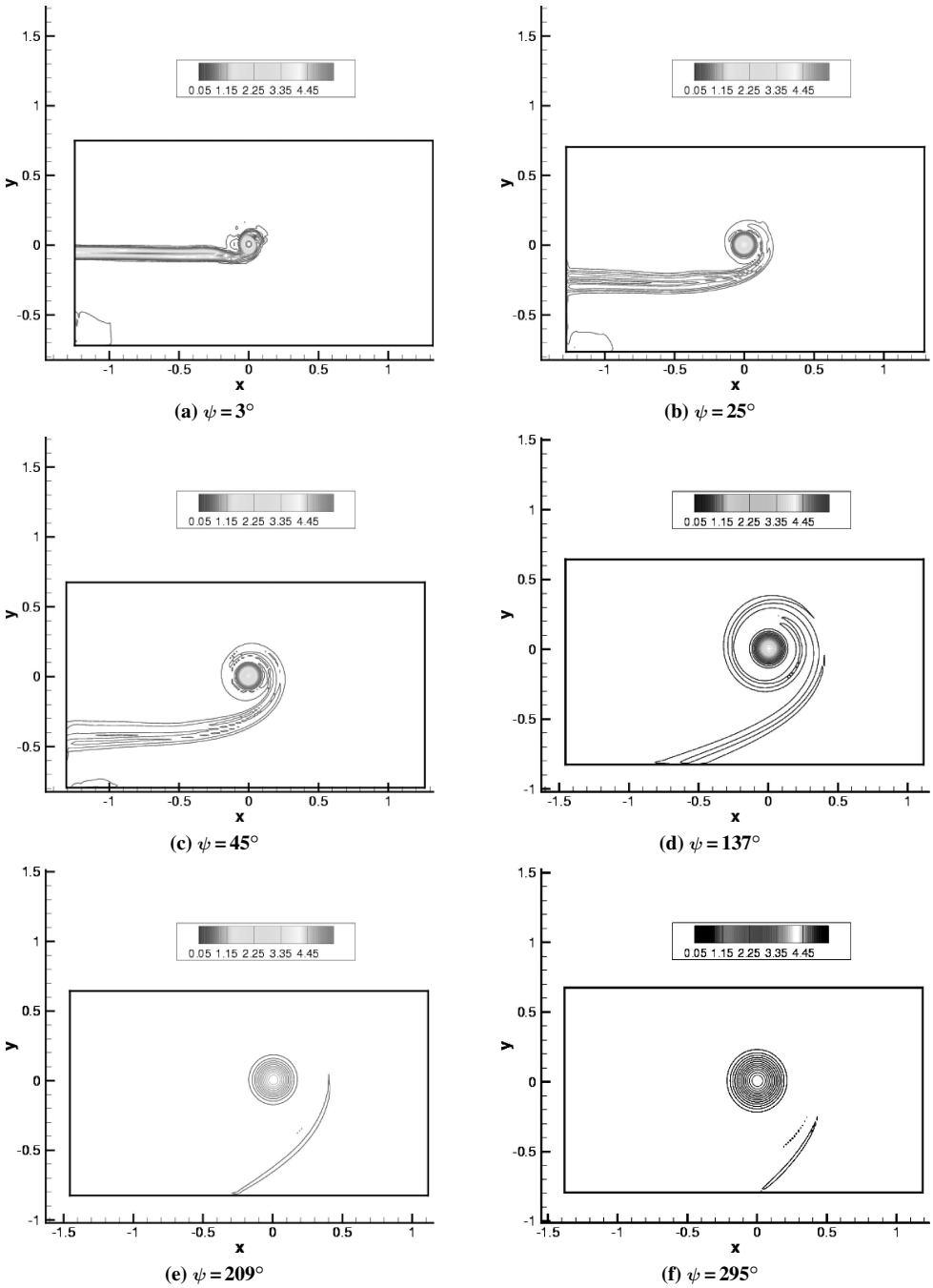
Fig. 1. Grid system for a single-bladed hovering rotor.

Numerical results

Figure 2 shows the vorticity magnitude along selected azimuthal planes in the overset vortex grid. It is evident that by a wake age of 25°, the vortex has rolled up completely, and the magnitude of the vorticity in the tip vortex is very much higher than that in the wake. Also, the vortex structure seems to have approached axisymmetry by the 25° wake age. Overall, the vorticity in the tip vortex diffuses at a much slower rate than the vorticity in the trailing vortex sheet, a result that can be attributed to the curvature induced stabilizing effects.

Figures 3 and 4 compare the computed peak swirl velocity and core radius with experiments as a function of the wake age. The third-order scheme performs poorly compared to the fifth-order scheme, suggesting non-optimal spatial discretization that causes increased numerical dissipation. While idealized tests (Ref. 14) have indicated that around 15 points are required across the core to effectively eliminate discretization errors, the current computation uses only 7 points in the initial wake ages (increasing to around 20 at later wake ages).

To study the effect of increased resolution, the original overset vortex grid was replaced by a refined grid (extending to a wake-age of 45°) that was 1.5 times as fine in the cross-stream directions. This did not



(a) $\psi = 3°$ (b) $\psi = 25°$

(c) $\psi = 45°$ (d) $\psi = 137°$

(e) $\psi = 209°$ (f) $\psi = 295°$

Fig. 2. Contours of vorticity magnitude (normalized by tip speed and chord) along selected azimuthal planes in the vortex grid.

change the velocity predictions by more than 3%, and hence the resolution appears to be sufficient in the cross-stream directions for this particular case. It has to be mentioned, that the alignment of the grid with the vortex axis can minimize errors because of non-orthogonality of flow features, but could affect the computation of the axial gradients. Therefore, the solution cannot be assumed to be rigorously grid converged unless the axial discretization is refined as well.

The rapid rate of decay of the vortex in the initial wake ages could be attributed to the turbulence ingested from the separating shear layer. At later wake ages, relaminarization effects cause the vortex to decay at a slower rate. This qualitative feature is well represented in the computations. Figure 5 compares the swirl velocity profiles at specific azimuthal locations.

Accurate experimental measurement of the axial velocity is difficult because the aperiodic movement of the vortex can make the detection of the peak value (which usually occurs at the centroid of the core) highly uncertain. The corrected values are currently unavailable for this experiment, and hence comparisons are made with just the measured values. A recent set of near-blade PIV based measurements (Ref. 33) on the same test rotor also indicate that the axial velocity measurements of Ref. 11 are not as reliable as the swirl velocity measurements. Figure 6 shows that the computations predict a much larger peak axial velocity deficit. Figure 7 compares the computed profiles at specific azimuthal locations.

Figure 8 shows the effect of the rotational correction in the turbulence model. As expected, the baseline S–A model with no rotational correction

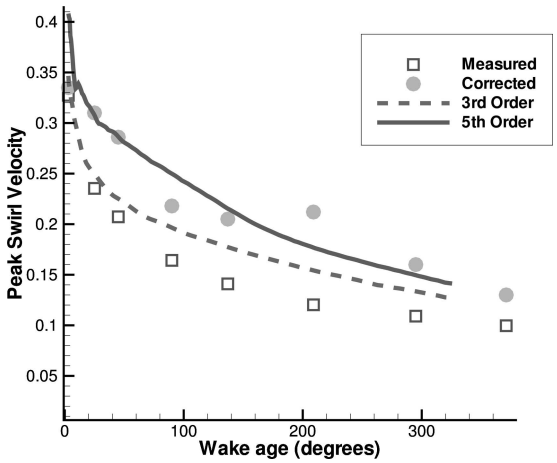


Fig. 3. Single hovering blade: Peak swirl velocity (normalized by tip-speed) as a function of wake age.

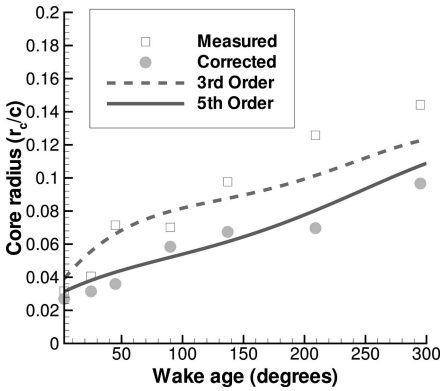


Fig. 4. Single hovering blade: Vortex core radius as a function of wake age. The computed core-radius is curve-fitted because of the discreteness of the grid.

proves to be extremely diffusive because it unphysically predicts maximum turbulence in the vortical core. On comparing the 3rd-order solution with rotational correction (Fig. 3) with the 5th-order solution with no correction (Fig. 8), it is apparent that turbulence modeling errors are much larger than numerical diffusion errors for this particular computational grid.

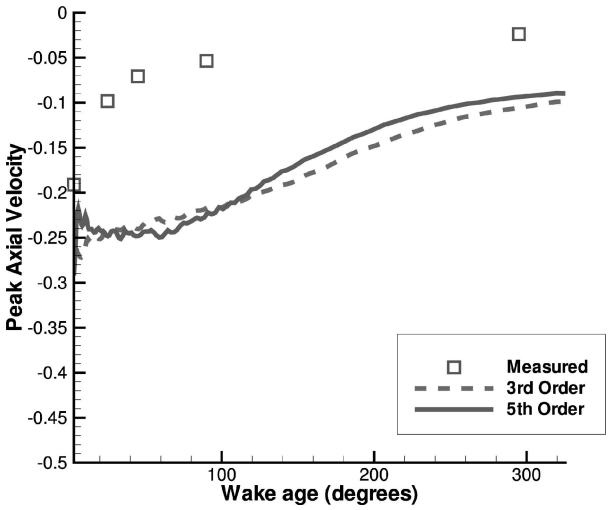


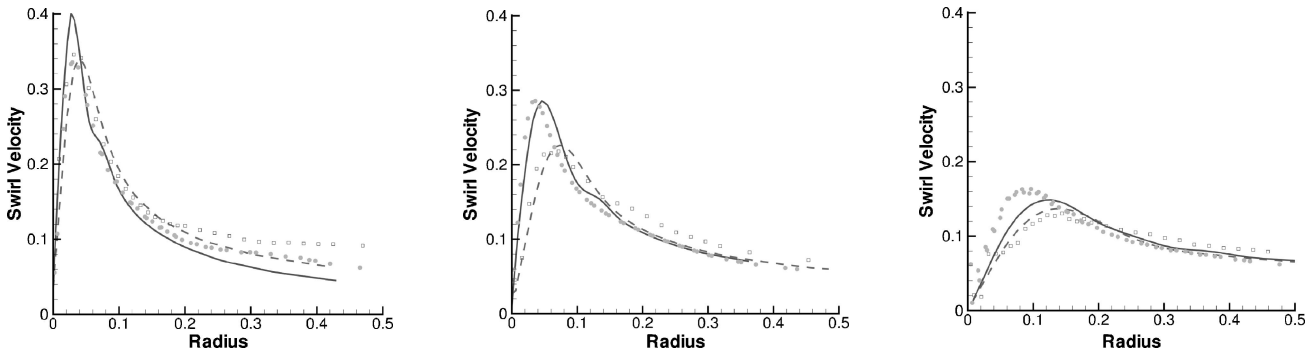
Fig. 6. Single hovering blade: Peak axial velocity deficit (normalized by tip-speed) as a function of wake age.

Two-Bladed Hovering Rotor

In the previous section, computations were performed on a single-bladed hovering rotor, with significant emphasis on the vortex velocity profiles. In this section, simulations will be performed on the two-bladed experimental setup of Caradonna and Tung (Ref. 12). This test case was chosen because it provides detailed blade surface pressure measurements and vortex trajectory data. This test case has also been previously validated using the Euler and RANS equations by several researchers (Refs. 24, 34, 35).

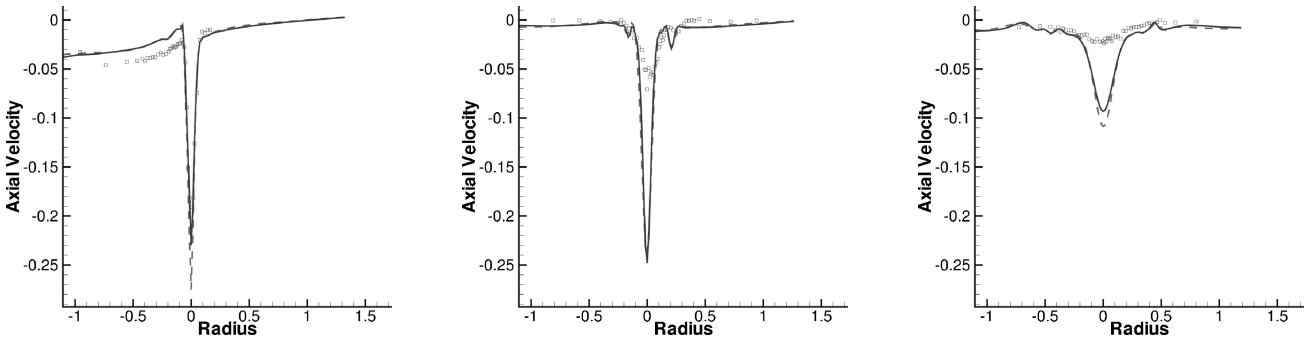
Experimental configuration

The experimental setup consisted of a two-bladed rigid rotor in a hover chamber. The blades were of a rectangular planform, and were untwisted with a radius of 1.143 m. The aspect ratio of the blade was 6. The blades used a NACA 0012 airfoil section along the entire span. A pre-cone angle of 0.5° was set for the blades. Of the different experiments conducted, Table 2 shows the cases that were chosen for validation in the present study. Of particular interest, and one that poses a challenge, is the 5° collective case. The associated thrust level is relatively low, causing the following blade to pass very close to the vortex. Accurate resolution of this phenomena could be critical in the prediction of hover performance.



(a) $\psi = 3°$ (b) $\psi = 45°$ (c) $\psi = 295°$

Fig. 5. Single hovering blade: Swirl velocity (normalized by tip-speed). Refer to Fig. 3 for legend.



(a) $\psi = 3°$ (b) $\psi = 90°$ (c) $\psi = 295°$

Fig. 7. Single hovering blade: Axial velocity (normalized by tip-speed). Refer Fig. 3 for legend.

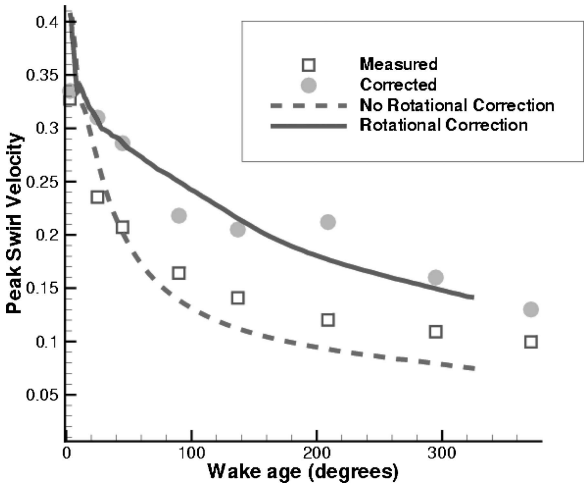


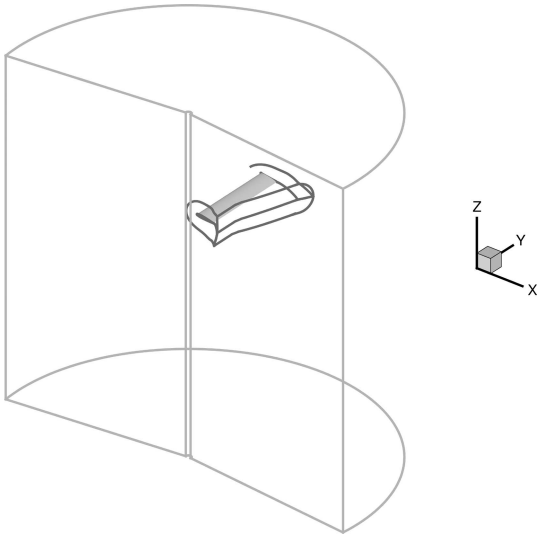
Fig. 8. Peak swirl velocity (normalized by tip-speed) predicted by S–A turbulence model with and without rotational correction. 5th-order inviscid differencing for both cases.

Table 2. Experimental test conditions (Ref. 12) and computed thrust for two-bladed rotor in hover. The pre-cone angle for all cases is 0.5°

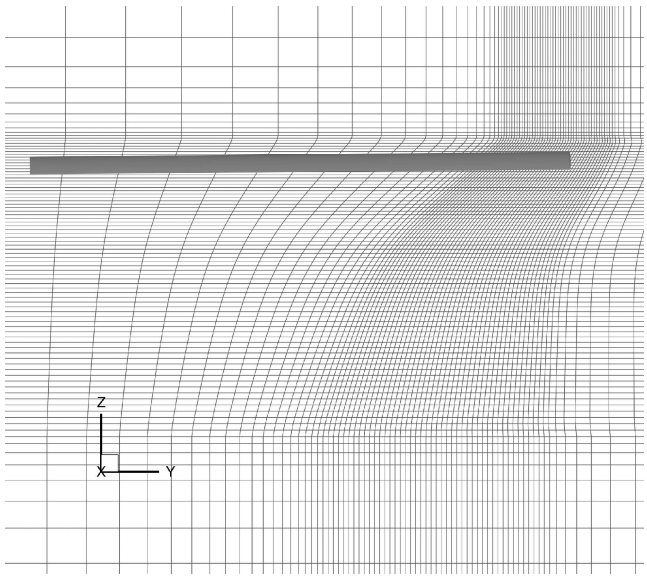
Tip Reynolds Number	Tip Mach Number	Collective Pitch	C_T	$C_{T\text{computed}}$
1.92×10^6	0.433	12°	0.00796	$0.0082 \pm 7 \times 10^{-5}$
1.96×10^6	0.439	8°	0.00459	$0.00473 \pm 1 \times 10^{-4}$
1.92×10^6	0.433	5°	0.00213	$0.00219 \pm 1.2 \times 10^{-4}$

Grid system

For the computations, a two grid system was used, the boundaries of which are shown in Fig. 9(a). A blade grid of C–O topology was embedded in a cylindrical background mesh. If the two blades are similar, the resulting flow field can be assumed to be symmetric, and hence periodic boundary conditions can be applied at the initial and final azimuthal planes, thus ensuring that only one blade (or half of the flow field) need be modeled. To preserve high-order of accuracy, a three level ghost cell overlap is used at the azimuthal boundary planes. The background grid is clustered in the region where the tip vortex is expected to evolve. This is seen in Fig. 9(b), where a single azimuthal plane is shown. The dimensions



(a) Blade and periodic background grid boundaries



(b) Azimuthal section of background grid (every other point in each direction)

Fig. 9. Grid system for two-bladed hovering rotor.

of the background grid are $97 \times 151 \times 199$ ($\approx$2.9 million points) in the azimuthal, radial, and vertical directions, respectively. The top and bottom boundaries are set at 1 and 2 rotor radii, respectively, and the outer boundary in the radial direction is at 2 rotor radii. The finest cross-stream cells in the background mesh correspond to squares of side $0.01c$.

The outer boundaries of the blade grid extend to approximately 1 chord-length from the surface. The corresponding overlap region in the background mesh is appropriately hole-cut with 4 fringe layers. For the 12° and 8° cases, the returning tip vortex is sufficiently far away (vertically) from the hole-cut region that it can be accurately resolved in the background grid (the "miss" distances are approximately around $0.5c$). For these cases, a blade grid of dimension $199 \times 129 \times 111$ ($\approx$2.9 million points) in the wrap-around, spanwise, and normal directions, respectively, was used. However, for the 5° case, the vortex was found to pass much closer to the blade surface, and the resulting interaction had to be resolved on the blade grid. For this reason, a much finer blade grid of dimensions $239 \times 169 \times 121$ ($\approx$4.9 million points) was used. Further discussion on this grid will be provided later in the paper. The normal spacing at the wall surface is set to $1 \times 10^{-5}c$, which results in $y^+ \approx O(1)$ at all blade surface locations.

Numerical results

For all the computations, the 5th-order WENO upwind scheme was used for discretization of the inviscid fluxes and second order central differencing was used for the viscous fluxes. The Spalart–Allmaras turbulence model with rotational correction was used for the RANS closure.

12° and 8° cases. Figure 10 shows the surface pressure distribution for the 12° collective case using the original grid system and a coarse grid system

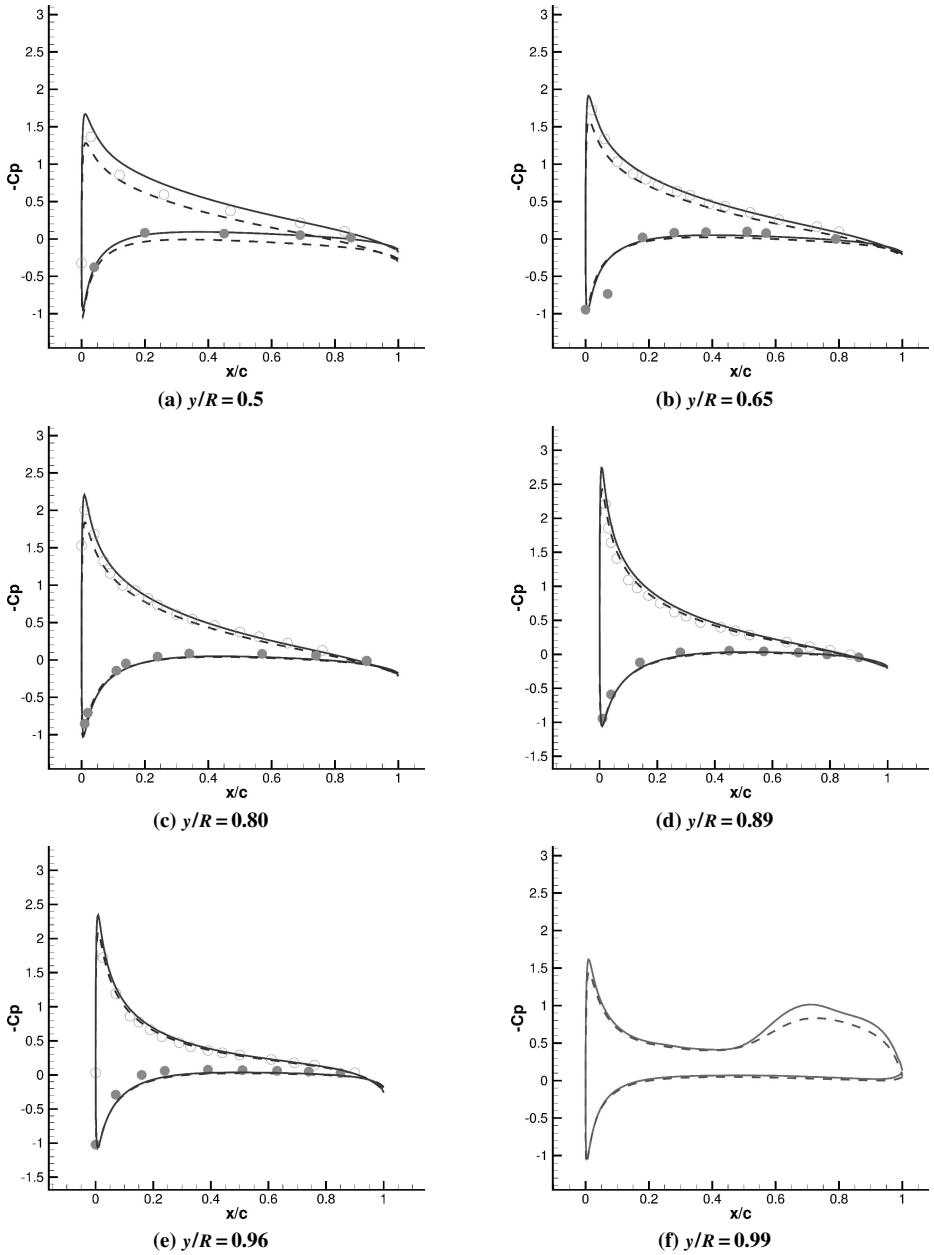


Fig. 10. Comparison of computed blade surface pressure coefficient using fine (solid lines) and coarse (dashed lines) grids with experiment (circles) for $\theta_0 = 12^\circ$.

that uses every other point in the cross-stream directions (resulting in one-fourth the number of total points). In these figures, the local streamwise velocity ($U(r)$) is used to normalize the pressure.

It is seen that the fine grid solution compares well with the experimental measurements at all the radial locations. The coarse grid solution is noticeably poor in the inboard stations, which is possibly a result of inadequate resolution in the spanwise direction. At the $y/R = 0.99$ station (experimental measurements not available here), the secondary suction peak from the presence of the tip vortex is evident. The fact that this peak is lower in magnitude in the coarse grid result can be attributed to a more diffuse resolution of the tip vortex.

Figure 11 shows the blade pressures for the 8° collective case (using the fine grid). Again, the results are of good quality except very close to the suction peak at the 80% and 89% span stations, where the agreement is off by slightly more than 5%.

A good criterion for the identification of flow field vortices is the q criterion (Ref. 36). q is the second invariant of the velocity gradient tensor (see nomenclature). It can be shown that q is positive in highly vortical regions and is negative in highly strained regions of the flow. This becomes a very useful tool in visualizing vortex dominated flow fields in that coherent structures can be identified with very little noise. Alternatively, if one relies on the total vorticity magnitude, features like boundary layers and wake sheets would be difficult to distinguish from tip vortices. Figure 12 shows iso-surfaces of the q variable (non-dimensionalized by tip speed and chord), clearly showing coherent multiple passes of the tip vortex.

Figure 13 shows the position of the vortex center compared to the curve-fitted experimental results. The accelerated downward convection after the first pass under the blade ($\psi = 180°$) is well represented. This sudden increase in downward convection is the result of the downwash

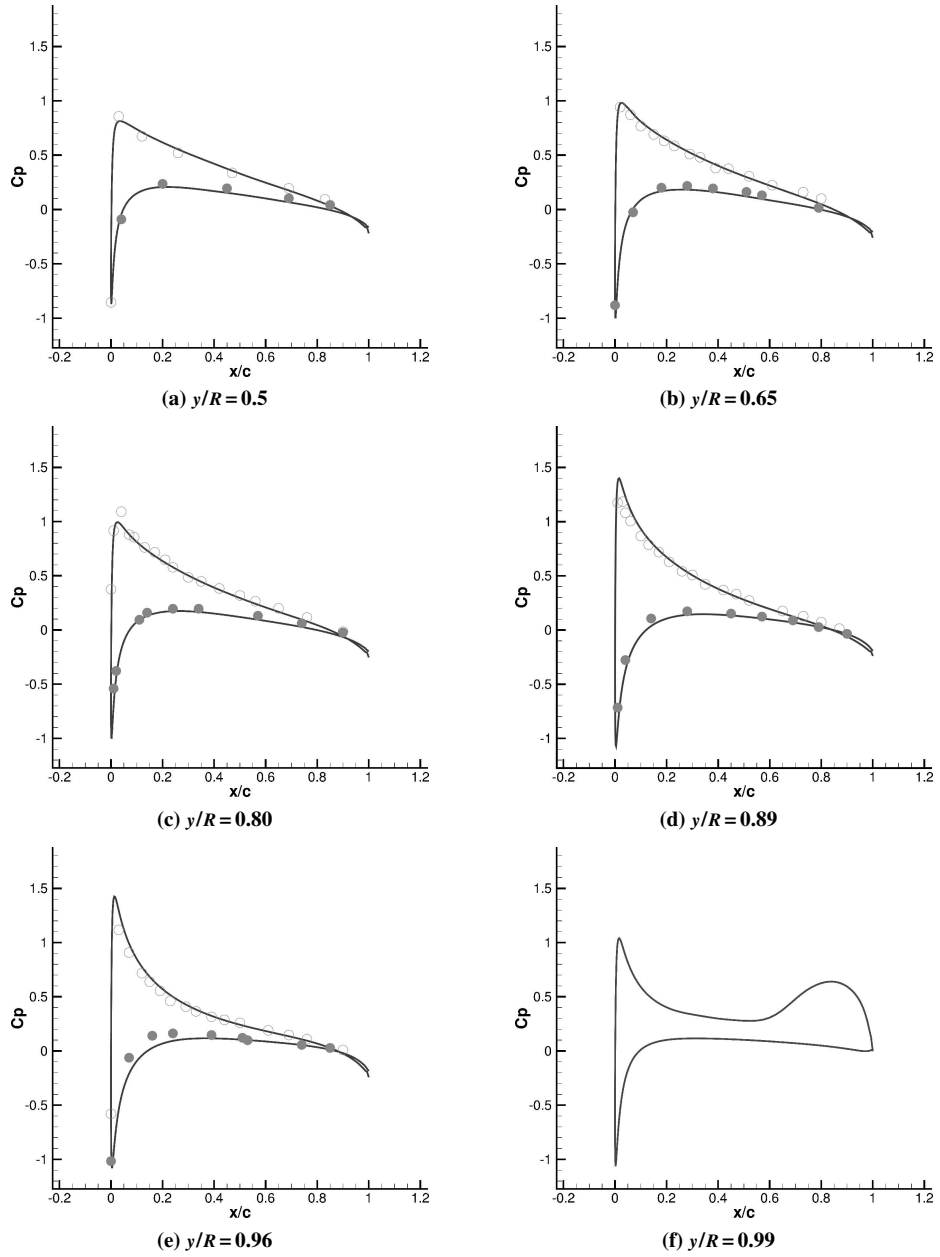


Fig. 11. Comparison of computed blade surface pressure (lines) coefficient with experiment (circles) for $\theta_0 = 8°$.

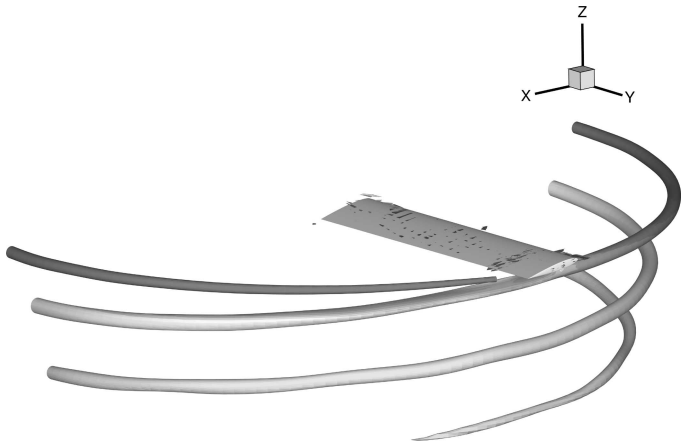
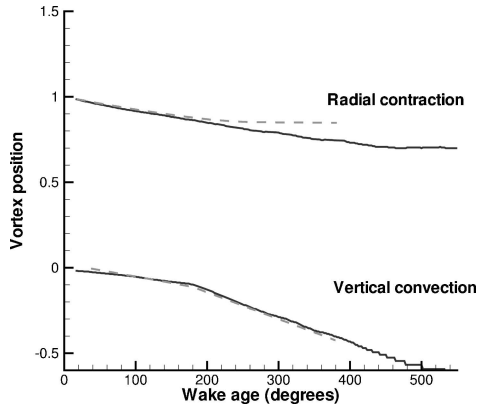
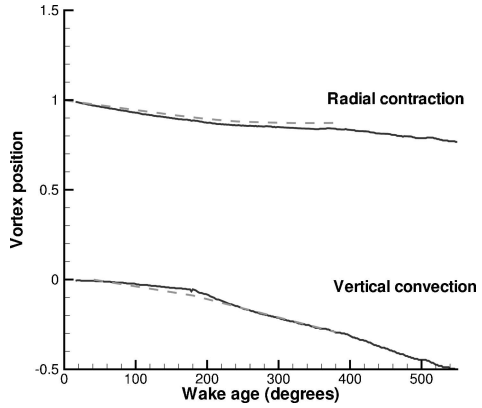


Fig. 12. Iso-surfaces of $q = 0.5$ (normalized by tip speed and chord) showing the tip vortex evolution for $\theta_0 = 8°$.



(a) $\theta_0 = 12°$



(b) $\theta_0 = 8°$

Fig. 13. Comparison of computed vortex center locations (solid lines) with curve-fitted experimental results (dashed lines).

from the blade and the evolving tip vortex on the first passage of the vortex. Overall, the comparison is reasonable, except for the radial contraction of the wake system at large wake ages. The experimental results are seen to converge at $0.86R$, whereas the computational results were found to converge close to the momentum theory predicted result of $R/\sqrt{2}$. The asymptotic contraction in the experiment hints at possible recirculation effects. However, the downward convection agrees well with experiments for all compared wake ages.

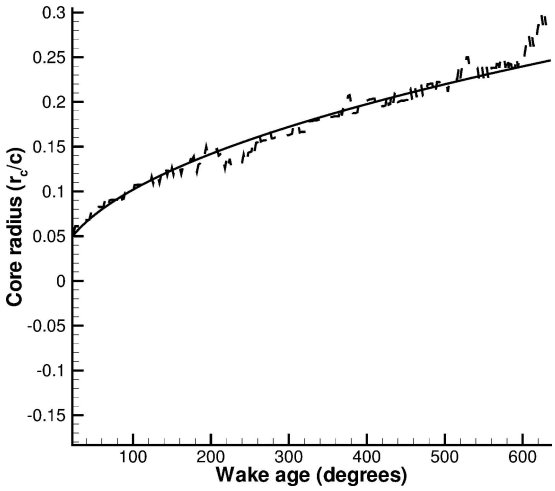
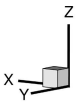


Fig. 14. Core radius evolution for $\theta_0 = 12°$. Dashed line: Distance between peak-to-peak grid points. Solid line: Curve fit.

The initial radius and growth rate of the vortical core for all the three collective pitch cases was similar, and for purposes of clarity the evolution characteristics of only the 12° case is shown in Fig. 14. The core grows by a factor of 4 after 3 blade passages ($\psi = 540°$). Because detailed experimental measurements of the vortex structure are not available, the effect of numerical diffusion and turbulence modeling inaccuracies cannot be determined. The detection algorithm returns the average distance between the grid points corresponding to the peak-to-peak swirl velocities for every wake age. The apparent oscillations are hence a result of the discreteness of the grid point distribution.

5° *case.* As mentioned earlier, in this case the tip vortex passes very close to the blade surface and as a result, had to be resolved within the blade mesh. The mesh points were refined off the surface in the normal direction, and in the spanwise direction near the region of the first vortex passage. Figure 15 shows a wrap-around plane of the blade grid. The clustering in the spanwise and normal directions are evident. Figure 16 compares the computed pressure distribution on the blade surface with the experimental measurements. Again, good agreement is achieved with the experimental data, especially at the 89% spanwise station, which is directly above the first pass of the vortex.



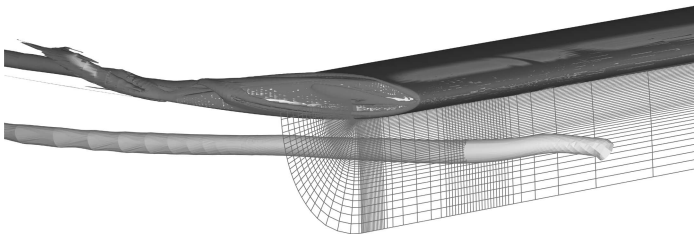
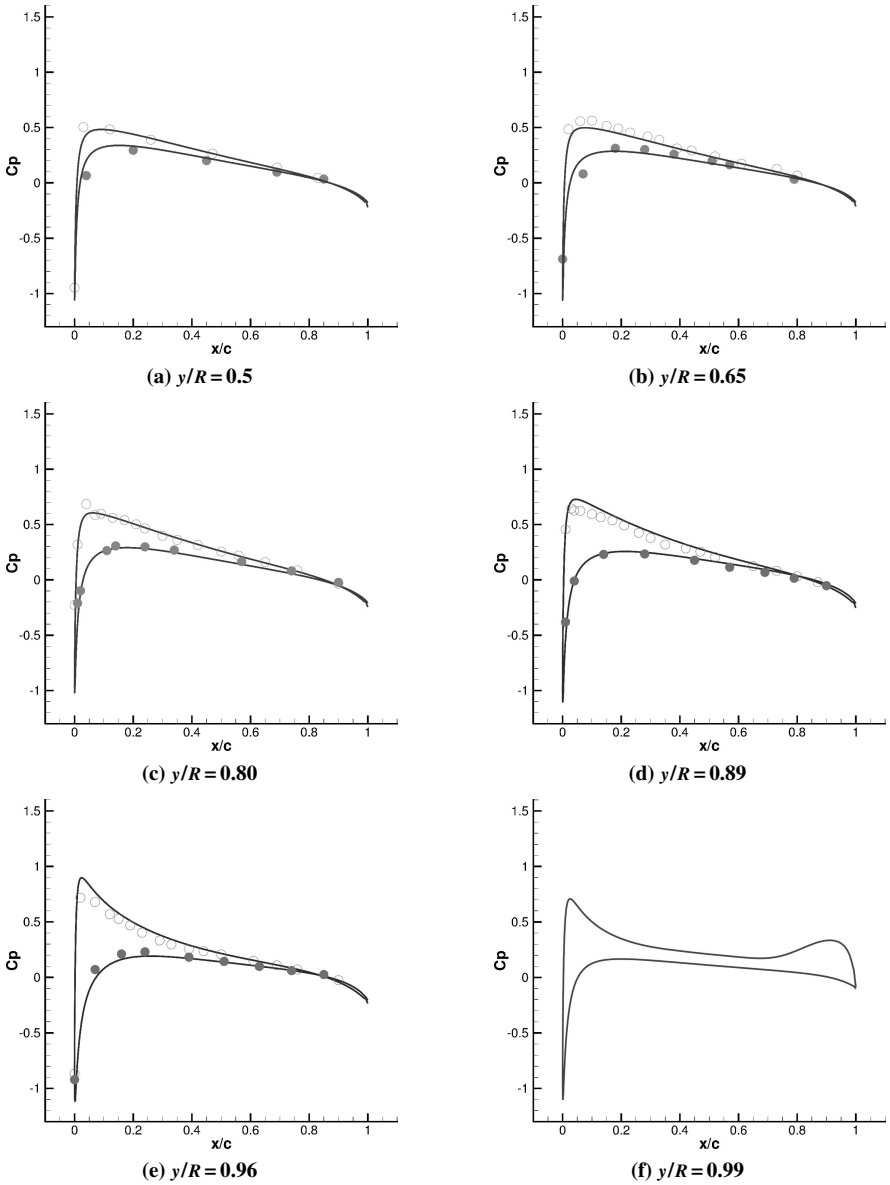


Fig. 15. Sample wrap-around plane of the blade grid (every other point in each direction) for $\theta_0 = 5°$. q iso-surfaces also shown.

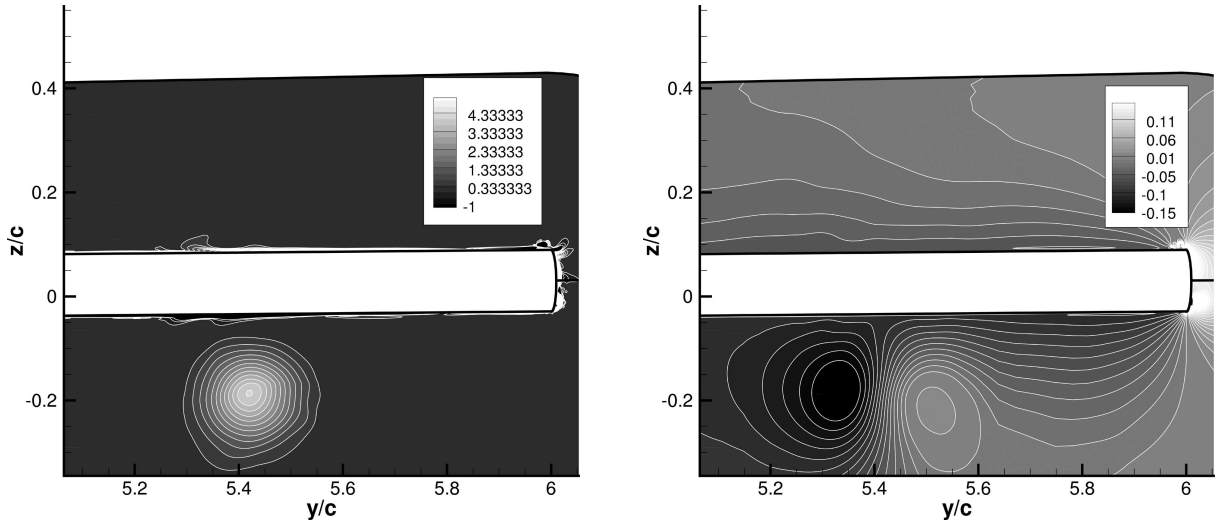


(a) $y/R = 0.5$ (b) $y/R = 0.65$

(c) $y/R = 0.80$ (d) $y/R = 0.89$

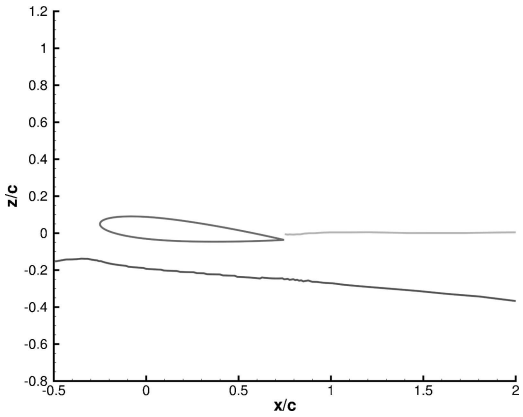
(e) $y/R = 0.96$ (f) $y/R = 0.99$

Fig. 16. Comparison of computed blade surface pressure coefficient (lines) with experiment (circles) for $\theta_0 = 5°$.

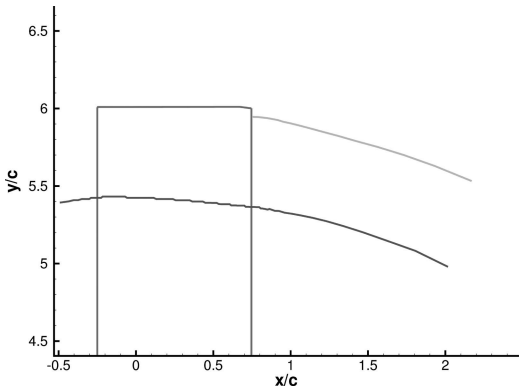


(a) Axial vorticity (normalized by tip speed and chord) **(b) Vertical velocity (normalized by tip speed)**

Fig. 17. First pass of vortex at quarter-chord section for $\theta_0 = 5°$ showing perpendicular blade vortex interaction.



(a) Side view



(b) Plan view

Fig. 18. Computed near-blade vortex positions for $\theta_0 = 5°$.

Further evidence of the interaction of the vortex with the blade and its strength can be seen from Fig. 17, in which the axial vorticity and vertical velocity are shown along a plane passing though the quarter chord position of the blade. The presence of the vortex is clearly seen to disturb the boundary layer vorticity (Fig. 17(a)). Figure 17(b) shows that the vortex induces a significant amount of vertical velocity on the blade surface, which in turn would affect the effective angle of attack of the section. The fact that good agreement is seen in Fig. 16(d) suggests that adequate resolution is achieved.

Figure 18 shows the tip vortex trajectory and the effect of the blade on the vortex during the first passage. As the tip vortex is formed and trailed off the blade surface, there is no downward convection in the initial stages. As the vortex nears the blade for its first pass, the downward convection tapers off. When it is directly under the leading edge of the blade, the presence of the blade downwash is seen to abruptly change the vortex position in such a way that the descent rate is increased very rapidly. Figure 19 compares the computed vortex position with curve-fitted experimental measurements. Again, reasonable agreement is achieved until slightly beyond the first pass, after which the experimental measurement reaches asymptotic contraction at $0.88R$, whereas the computed wake contracts further.

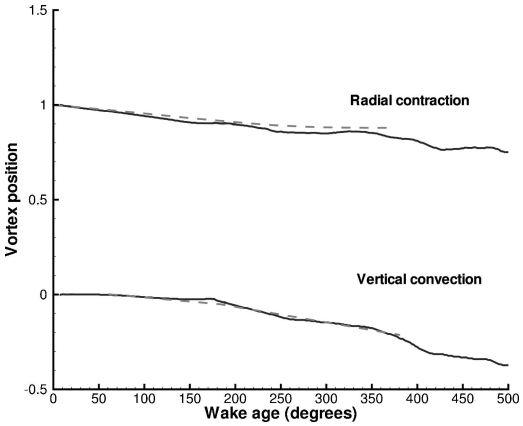


Fig. 19. Comparison of computed vortex center locations (solid lines) with curve-fitted experimental results (dashed lines) for $\theta_0 = 5°$.

Conclusions

Highly accurate computation of the wake system is imperative in the design of efficient helicopters. The RANS equations represent the highest level of fluid dynamic approximation that is currently feasible in the simulation of high Reynolds number flows. However, practical RANS simulations of tip vortex formation and evolution suffer from 1) numerical diffusion, which arises primarily from inaccuracies in the discretization of the convective term in the NS equations, and 2) inaccurate turbulence modeling, which is a result of the unavailability of a perfect closure model to the RANS equations.

In the present work, the error from numerical diffusion was reduced by the use of proper grid resolution (by means of grid clustering and overset meshes) and high-order accurate numerical schemes. None of the available turbulence models are accurate or efficient enough to completely describe the complex tip vortex flow field, but reliable solutions can be obtained by adding a simple rotational correction to the production term in the Spalart–Allmaras turbulence model. For the first time, an attempt has been made at detailed validation of the vortex structure trailed from rotating blades. The following are the specific conclusions noted from this study:

1) The use of a refined overset mesh, high-order schemes and turbulence model modification was seen to be essential to accurately resolve the evolving tip vortex.

2) For the computations on the single-bladed rotor, fair agreement was achieved with the experimental swirl velocity measurements up to 300° of wake evolution behind the trailing edge.

3) For the two-bladed rotor, the tip vortex could be tracked up to four wake passages behind the trailing edge. Good correlation was observed with experimental surface pressure measurements and vortex trajectory data.

4) With the use of appropriately refined meshes, the interaction of the blade with the first vortex passage was found to be well represented.

Taken together, the above set of reference data represent a spectrum of validations that establish the fidelity of the methodology in predicting hovering rotor flow fields in an efficient manner. There is however, wide scope for improvement and avenues for future work. A primary suggestion for the continuation of this work would be conduct a more rigorous grid convergence study. Advanced turbulence models such as the Second Moment Closure models (Ref. 19) can be used to better represent curvature and anisotropic effects. Application of the current methodology to more realistic hovering rotors would require slightly different mesh topologies, for instance, to resolve the inboard vortex sheet.

Appendix

To obtain an estimate of the computational cost, several of the test cases were evaluated for CPU time. On average, approximately 11 seconds of CPU time was required per iteration per million points on a single Intel Xeon processor with 2 GB of RAM and a clock speed of 3.33 GHz. Typically, around 8,000 iterations were found to be sufficient to achieve

Table 3. Typical break down of CPU time per iteration

Procedure	Time per Iteration (%)
Inviscid RHS	35
Viscous RHS	26
LHS and Inversion	20
Turbulence model	18
BC and interpolation	<1

a converged solution for the cases presented in this paper, resulting in a computational cost of less than 100 CPU hours. Table 3 breaks down the time per iteration for a representative overset mesh computation.

Acknowledgments

This work was conducted at the Alfred Gessow Rotorcraft Center at the University of Maryland and has been supported by the National Rotorcraft Technology Center under the Rotorcraft Centers of Excellence program.

References

[1]Beddoes, T. S., "A Wake Model for High Resolution Airloads," International Conference on Rotorcraft Basic Research Proceedings, Research Triangle Park, NC, 1985.

[2]Egolf, T. A., "Helicopter Free Wake Prediction of Complex Wake Structures Under Blade-Vortex Interaction Operating Conditions," American Helicopter Society 44th Annual Forum Proceedings, Washington, DC, June 16–18, 1988.

[3]Quackenbush, T., Wachpress, D., and Boschitsch, A., "Computation of Rotor Aerodynamic Loads with a Constant Vorticity Contour Free Wake Model," *Journal of Aircraft*, Vol. 32, (5), 1995, pp. 911–920.

[4]Bhagwat, M. J., and Leishman, J. G., "Stability, Consistency and Convergence of Time Marching Free-Vortex Rotor Wake Algorithms," *Journal of the American Helicopter Society*, Vol. 46, (1), 2001, pp. 59–71.

[5]Bagai, A., and Leishman, J. G., "Rotor Free-Wake Modeling using a Relaxation Technique—Including Comparisons with Experimental Data," *Journal of the American Helicopter Society*, Vol. 40, (2), April 1995, pp. 29–41.

[6]Steinhoff, J., and Ramachandran, K., "Free-Wake Analysis of Compressible Rotor Flows," *AIAA Journal*, Vol. 28, (3), 1990, pp. 427–431.

[7]Sitaraman, J., Baeder, J. D., and Chopra, I., "Validation of UH-60 Rotor Blade Aerodynamic Characteristics Using CFD," American Helicopter Society 59th Annual Forum Proceedings, Phoenix, AZ, May 6–8, 2003.

[8]Duque, E. P. N., and Srinivasan, G. R., "Numerical Simulation of a Hovering Rotor Using Embedded Grids," American Helicopter Society 48th Annual Forum Proceedings, Washington, DC, June 3–5, 1992.

[9]Ahmad, J., and Strawn, R. C., "Hovering Rotor and Wake Calculations with an Overset Grid Navier–Stokes Solver," American Helicopter Society 55th Annual Forum Proceedings, Alexandria, VA, May 25–27, 1999.

[10]Potsdam, M., Yeo, H., and Johnson, W., "Rotor Airloads Prediction using Loose Aerodynamic/Structural Coupling," American Helicopter Society 60th Annual Forum Proceedings, Baltimore, MD, June 7–10, 2004.

[11]Martin, P. B., Pugliese, G. J., and Leishman, J. G., "High Resolution Vortex Measurements in the Wake of a Hovering Rotor," American Helicopter Society 57th Annual Forum Proceedings, Phoenix, AZ, May 9–11, 2001.

[12]Caradonna, F. X., and Tung, C., "Experimental and Analytical Studies of a Model Helicopter Rotor in Hover," NASA TM 81232, 1981.

[13]Hirsch, C., *Numerical Computation of Internal and External Flows*, Volume 1, Wiley, Chichester, England, 1988.

[14]Wake, B. E., and Choi, D., "Investigation of High-Order Upwinded Differencing for Vortex Convection," *AIAA Journal*, Vol. 34, (2), 1996, pp. 332–337.

[15]Spalart, P. R., "Trends in Turbulence Treatments," AIAA Paper 2000-2306, AIAA Fluids 2000 Conference and Exhibit Proceedings, Denver, CO, June 19–22, 2000.

[16]Egolf, T. A., "Recent Rotor Wake Simulation and Modeling Studies at United Technologies Corporation," AIAA Paper 2000-0115, 38th AIAA Aerospace Sciences Meeting and Exhibit Proceedings, Reno, NV, January 10–13, 2000.

[17]Wilcox, D. C., *Turbulence Modeling for CFD*, DCW Industries Inc., La Canada, CA, 1993.

[18]Spalart, P. R., and Allmaras, S. R., "A One-Equation Turbulence Model for Aerodynamic Flows," AIAA Paper 92-0439, 30th AIAA Aerospace Sciences Meeting and Exhibit Proceedings, Reno, NV, January 6–9, 1992.

[19]Pope, S. B., "A More General Eddy-Viscosity Hypothesis," *Journal of Fluid Mechanics*, Vol. 72, 1975, pp. 331–340.

[20]Jakirlic, S., Hanjalic, K., and Tropea, C., "Modeling Rotating and Swirling Turbulent Flows: A Perpetual Challenge," *AIAA Journal*, Vol. 40, (10), 2002, pp. 1984–1996.

[21]Spalart, P. R., and Shur, M., "On the Sensitization of Turbulence Models to Rotation and Curvature," *Aerospace Science and Technology*, Vol. 1, (5), 1997, pp. 297–302.

[22]Pettersson-Reif, B. A., Durbin, P. A., and Ooi, A., "Modeling Rotational Effects in Eddy-Viscosity Closures," *International Journal of Heat and Fluid Flow*, Vol. 20, (6), 1999, pp. 563–573.

[23]Devenport, W. J., Rife, M. C., Liapis, S. I., and Follin, G. J., "The Structure and Development of a Wing-Tip Vortex," *Journal of Fluid Mechanics*, Vol. 312, 1996, pp. 71–104.

[24]Leishman, J. G., "Measurements of the Aperiodic Wake of a Hovering Rotor," *Experiments in Fluids*, Vol. 25, 1998, pp. 352–361.

[25]Srinivasan, G. R., and Baeder, J. D., "TURNS: A Free-wake Euler/Navier–Stokes Numerical Method for Helicopter Rotors," *AIAA Journal*, Vol. 31, (5), 1993, pp. 959–962.

[26]Shu, C-W., "Essentially Non-Oscillatory and Weighted Essentially Non-Oscillatory Schemes for Hyperbolic Conservation Laws," ICASE Report 1997-65, 1997.

[27]Chow, J. S., Zilliac, G. G., and Bradshaw, P., "Mean and Turbulence Measurements in the Near Field of a Wingtip Vortex," *AIAA Journal*, Vol. 35, (10), 1997, pp. 1561–1567.

[28]Jacquin, L., and Pantano, C., "On the Persistence of Trailing Vortices," *Journal of Fluid Mechanics*, Vol. 471, 2002, pp. 159–168.

[29]Dacles-Mariani, J., Chow, J. S., Zilliac, G. G., and Bradshaw, P., "Numerical/Experimental Study of a Wingtip Vortex in a Near Field," *AIAA Journal*, Vol. 33, (9), 1995, pp. 1561–1568.

[30]Heyes, A., Jones, R., and Smith, D., "Wandering of Wing Tip Vortices," 12th International Symposium on Applications of Laser Techniques to Fluid Mechanics Proceedings, Lisbon, July 12–15, 2004.

[31]Martin, P. B., "Measurements of the Trailing Vortex Formation, Structure and Evolution in the Wake of a Hovering Rotor," Ph.D. Dissertation, Department of Aerospace Engineering, University of Maryland, 2001.

[32]Landgrebe, A. J., "The Wake Geometry of a Hovering Rotor and Its Influence in Rotor Performance," *Journal of the American Helicopter Society*, Vol. 17, (4), 1972, pp. 3–15.

[33]Duraisamy, K., Ramasamy, M., Baeder, J. D., and Leishman, J. G., "Computational/Experimental Study of Hovering Rotor Tip Vortex Formation," American Helicopter Society 62nd Annual Forum Proceedings, Phoenix, AZ, May 9–11, 2006.

[34]Hariharan, N., and Sankar, L., "High Order Essentially Nonoscillatory Schemes for Rotary-Wing Wake Computations," *Journal of Aircraft*, Vol. 41, (2), 2004, pp. 258–267.

[35]Usta, E., "Application of a Symmetric Total Variation Diminishing Scheme To Aerodynamics of Rotors," Ph.D. Dissertation, Department of Aerospace Engineering, Georgia Institute of Technology, 2002.

[36]Dubief, Y., and Delcayre, F., "On Coherent-Vortex Identification in Turbulence," *Journal of Turbulence (Online)*, Vol. 1, January 2000.